



**NANYANG
TECHNOLOGICAL
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NTU Academy for Professional
and Continuing Education

Building with AI

Quick Prototyping : Build a working web app from a single written brief

Module 5

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The Builder's Ladder — you are here

Each rung builds on the one below, and today we take the first.

- **Rung 1 · Prototype** — "I can build" — today
- **Rung 2 · Scaffold** — "I can choose what to build and start it properly"
- **Rung 3 · Practice** — "I can build responsibly"

What you'll learn today

What a **tech stack** is — the four layers every app shares

How AI platforms **collapse the stack** — and build from one instruction

How to write a **five-part brief** that gets good work out of an AI

How to **scope**: one job first — the most lovable one, if it's for others

How to stay in control: the **build loop** — Brief → Build → Poke → Patch

Get set up — do this now

- Open **aistudio.google.com** — sign in with any Google account (free)
- Click **Build** in the left menu (may be labelled "Vibe code" — same thing)
- Using Safari? Switch to Chrome, Edge, or Firefox

SEE IT · 1 OF 6 — WHAT YOU'RE ABOUT TO BUILD

See it

Section 01 · What you're about to build

The app: help me weigh a real decision

"Stay in my current role — or take the new job offer?"

We will watch it work first; then you will build the same app yourselves.

BUILD IT · 2 OF 6 — EVERYONE, TOGETHER

Build it

Section 02 · Everyone, together, from one instruction

Paste and run — together

- Copy the **build prompt** I've just shared
- Paste it into the AI Studio prompt box

While it generates: plan → generate → assemble

The AI **plans** first — watch it list the five sections it intends to build

Then it **writes code** across multiple files — a pause here is normal

Then it **assembles** a live preview — total: 1–3 minutes

Poke it

- Change a few scores and weights — **watch the bars move**
- **Refresh** the page — is your data still there?
- Click **Start Fresh** — everything resets

From Python scripts to a shipped web app

THE TECH STACK · 3 OF 6 — WHAT DID YOU BUILD

The tech stack

Section 03 · What did you just build?

Key concept: a tech stack is a team of four

A **tech stack** is the set of technologies that work together to make an app run.

Front end · Back end · Database · Services

Your app is a restaurant

Restaurant	The layer	In the app you built
Dining room	Front end — what users see	The rows, dropdowns, the two bars
Kitchen	Back end — the thinking	Weight × score, summed → the Verdict
Storage room	Database — the memory	Your data survived the refresh
Suppliers	Services — outside help	Empty so far — deliberately

The stack AI Studio scaffolded for your app

Layer	In the app you built	When you need more
Front end	React + TypeScript · Tailwind CSS	—
Back end	The scoring logic — TypeScript, in your browser	Node.js runtime — secure API calls
Database	localStorage — no accounts, works offline	Firebase — auto-provisioned
Services	Empty — excluded by the brief	Gemini API — key kept server-side

Source: the generated app's own stack report · Google AI Studio Build mode docs, 2026.

Why the layers matter: name the layer, fix the right thing

You see	The layer	Tell the AI
Button does nothing	Front end	"The [X] button doesn't respond — fix it"
Numbers wrong, bars stuck	Back end	"Every score change must update the bars"
Data gone after refresh	Database	"Save everything to localStorage"

Key concept: the stack has collapsed

Assembling these layers once took a team **weeks**. AI platforms now generate all four from a paragraph — in **minutes**.

*The scarce skill is no longer assembling the layers; it is **directing** the AI that does.*

THE BUILDER'S METHOD · 4 OF 6 — HOW TO DIRECT AN AI

The Builder's Method

Section 04 · How to direct an AI builder

Key concept: the prompt you pasted is a brief

You have briefed contractors, vendors, and new hires throughout your career.

That same skill is what builds software now.

A good brief has five parts

Part	Skipping it costs you
1 · What it is — one sentence	A meandering app — several half-things
2 · Who it's for — and the problem	A generic app nobody chooses
3 · What it must do — numbered, specific	The AI decides for you — often wrongly
4 · What it looks like — one real example	Emptiness you can't judge
5 · What's out — stated exclusions	Features you never asked for

Key concept: what you don't specify, the AI decides

"Users can manage their criteria" leaves the AI to make **five design guesses**

"An input with an Add button; each row has a delete button and a weight dropdown (1–5)..." leaves it **none**

Source: Lovable prompting documentation, 2026 – "it does not infer intent."

Who is it for? That decides what you build first

Building for **yourself**: build the job you need, and let appearance wait

Building for **others**: find the feature that makes them **choose** it, and build that first

In this app, that feature is **the two bars**: the whole decision visible at a glance

Key concept: one job first

*Do not start feature two until feature one works **end to end**.*

Key concept: the build loop

Brief → **Build** → **Poke** → **Patch** — then poke again.

Brief — the whole instruction, written first

Build — the AI generates the app

Poke — test before you trust

Patch — one change, aimed at a named layer

Poke before you trust — the evidence

In a 2025 controlled trial, experienced developers using AI tools felt **~20% faster** — and measured **~19% slower**.

*The only reliable evidence is what you find when you **poke the output**.*

Source: METR randomised controlled trial, 2025.

IMPROVE IT · 5 OF 6 — TWO PATCHES

Improve it

Section 05 · Two patches: a better feature, then AI

Improve a feature: highlight the winner

- Paste the prompt into the app's chat panel and apply it
- When it finishes, test the app again — confirm the verdict still works

What would this app need AI for?

Everything it does so far is arithmetic — weights, totals, a comparison. **No AI required.**

AI earns its place where **judgment** is needed, not calculation

In a decision app, the judgment gap is clear: **the factors you didn't think of**

Make it an AI app: ask for what you missed

- Paste the prompt and apply it, then click your new button
- The AI reads your decision and recommends a factor you may have missed

Your app just called the AI that built it

ZOOM OUT · 6 OF 6 — THE MAP AND THE NEXT RUNG

Zoom out

Section 06 · What you just did has a name

Today you ran the Builder's Method

Brief → Build → Poke → Patch.

One step was done for you: ***the brief.***

Key concept: from vibe coding to spec-first building

Vibe coding — type a prompt and accept whatever comes back

Spec-first — write down what to build **before** the AI builds it

Today's brief was your first spec. The industry is moving the same way.

Source: spec-driven development — GitHub Spec-Kit, 2025–2026.

The PRD: a brief that grew up

Your brief (today)	The PRD adds
What it is · who it's for	The problem, evidenced — who has it, what it costs
What it must do	Numbered, testable requirements
What's out	Non-goals — the AI scope-creeps without them
—	Success criteria — how do we know it worked?

Make a PRD with AI — three prompts

1. **Describe the problem** — "I struggle with... I have tried... but..."
2. **Discuss the scope** — "help me decide the smallest lovable version"
3. **Ask for the draft** — "write a PRD: problem, user, requirements, non-goals — under 500 words"

The AI writes the draft; ***the decisions stay with you.***

The PRD guides the AI — step by step

Small app (3–5 features) → paste the whole PRD as **one prompt**

Bigger app → build **one requirement at a time**: build → poke → next

Every step is the **loop you already know**

Key concept: start from a scaffold, not a blank page

A **scaffold** = a proven starting point — stack chosen, structure set

Each scaffold comes with a **ready-made build prompt** — an expert's brief

Copy it, paste it into your AI tool, and layer **your** brief on top

Scaffold Studio — pick a shape, copy its prompt

scaffolding-applications.vercel.app

Browse **five shapes**: AI apps · Data apps · Websites · No-UI apps · Teaching repos

Open a scaffold, read what each part of its stack does, and **copy its ready-made prompt**

Share your app

- Click **Share** (top right), copy the link, and post it for the class
- Anyone you share with can **see the code and copy it** — within this class, that is a feature

You leave with three things

1. **A tool** — a working decision app you built, ready for a real decision tonight
2. **A method** — a clear brief, one job first, and testing before trusting
3. **A rung** — from analysing data to **shipping tools**, by directing AI

Your next build — this week

- Put a **real decision** of yours into the app tonight — that is what it is for
- Write your **own five-part brief** — for a real problem you have
- Pick its shape at **scaffolding-applications.vercel.app** — copy the scaffold prompt
- When the idea grows: draft a **PRD with the three prompts**, build step by step
- The **study guide** carries everything: the catalog, the patches, the PRD recipe

Version Management

Month/Year	Version	Changes
May 2026	0.0	New Module Design
July 2026	1.0	Refreshed Content and latest Information added